

Hassan Mohamed

MERN Stack Developer

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Summary

Full-Stack MERN Developer building production-grade web applications end to end with MongoDB, Express.js, React and Node.js. Experienced in designing normalized and document-based data models, building secure REST and real-time APIs with JWT and role-based access control, and delivering responsive, performant React frontends backed by Redux Toolkit. Comfortable across the whole delivery cycle — schema design, API architecture, state management, caching, testing, containerization and deployment. Always looking for a new challenging project to work on.

Projects

MULTI-VENDOR E-COMMERCE MARKETPLACE

React JS - Redux Toolkit - Node.js - Express - MongoDB - Mongoose - JWT - Stripe - Cloudinary - Redis - Docker

- Architected a full multi-vendor marketplace where independent sellers manage their own storefronts, inventory and order pipelines under a single platform, with an admin layer overseeing vendors, commissions and payouts.
- Designed the MongoDB data model with Mongoose (users, vendors, products, variants, carts, orders, reviews, payouts), using compound indexes and aggregation pipelines for faceted search, category filtering and sales reporting.
- Built a secure REST API on Express with JWT access/refresh tokens, bcrypt password hashing, role-based access control (buyer / vendor / admin), rate limiting, Helmet, request validation and centralized error handling.
- Integrated Stripe for card payments plus cash-on-delivery, handling webhooks for payment confirmation, refunds and failed-charge reconciliation to keep order state consistent.
- Implemented product image uploads through Multer and Cloudinary with server-side optimization, and cached hot catalog queries in Redis to cut repeated database reads.
- Managed frontend state with Redux Toolkit and RTK Query — cart, wishlist, filters, pagination and optimistic updates — and built vendor and admin dashboards with sales charts and order management.
- Containerized the API and frontend with Docker Compose and deployed with environment-based configuration and CI checks on every push.

REAL-TIME COLLABORATION & PROJECT MANAGEMENT PLATFORM

React JS - Node.js - Express - MongoDB - Socket.IO - Redux Toolkit - JWT - Tailwind CSS

- Built a Trello-style team workspace supporting organizations, projects, kanban boards, task cards, checklists, labels, due dates, attachments and threaded comments.
- Implemented real-time collaboration with Socket.IO — live card drag-and-drop across clients, presence indicators, typing indicators and instant notification delivery — using room-based event scoping per board.
- Designed a granular permission system (owner / admin / member / viewer) enforced at both the API layer and the UI, with invitation flows via email tokens.
- Modeled boards and ordered card lists in MongoDB using fractional position keys, allowing reordering without rewriting entire collections.
- Added a full activity feed and audit trail recording every mutation, plus in-app and email notifications through Nodemailer with user-configurable preferences.
- Built the frontend in React with Tailwind CSS and dnd-kit for accessible drag-and-drop, with Redux Toolkit slices synchronized against incoming socket events.

SOCIAL MEDIA NETWORK WITH LIVE MESSAGING

React JS - Node.js - Express - MongoDB - Socket.IO - Redux - Cloudinary - JWT

- Developed a complete social platform with profiles, follow/unfollow graphs, posts with image and video media, likes, nested comments, reposts, hashtags and full-text search.
- Engineered a personalized feed using MongoDB aggregation pipelines that blend followed accounts, engagement signals and recency, with cursor-based infinite scroll pagination.
- Built one-to-one and group chat over Socket.IO with message persistence, delivery and read receipts, online status, unread counters and typing indicators.
- Implemented media handling through Cloudinary with client-side compression, lazy loading and responsive image delivery to keep the feed fast on mobile connections.

- Added moderation tooling — report queues, content flagging, blocking and an admin dashboard for user and content management.
- Secured the platform with JWT refresh-token rotation, input sanitization against NoSQL injection and XSS, and per-route rate limiting.

LEARNING MANAGEMENT SYSTEM (LMS) WITH VIDEO STREAMING

Next.js - React - Node.js - Express - MongoDB - Stripe - FFmpeg - JWT - Tailwind CSS

- Built an end-to-end online course platform serving three roles — students, instructors and administrators — each with a dedicated dashboard and permission scope.
- Created an instructor course builder supporting sections, lessons, video uploads, downloadable resources, quizzes with auto-grading, and draft/publish workflows.
- Implemented adaptive video delivery by transcoding uploads with FFmpeg into HLS renditions, with signed URLs to prevent unauthorized access to paid content.
- Tracked granular student progress per lesson, resumable playback positions, quiz scores and automatic certificate generation on course completion.
- Integrated Stripe Checkout for one-off purchases and subscriptions, with coupon codes, instructor revenue splits and webhook-driven enrollment provisioning.
- Used Next.js server-side rendering for course and landing pages to improve SEO and first-paint performance, while keeping the authenticated app client-rendered.

RESTAURANT ORDERING & DELIVERY SYSTEM

React JS - Node.js - Express - MongoDB - Socket.IO - Redux Toolkit - Stripe - Google Maps API

- Developed a three-sided food ordering system connecting customers, restaurant staff and delivery riders through a shared real-time order lifecycle.
- Built dynamic menu management with categories, item variants, add-ons, stock availability and scheduled pricing, all editable from a restaurant dashboard.
- Implemented live order tracking with Socket.IO, pushing status transitions (placed, accepted, preparing, out for delivery, delivered) instantly to every connected party.
- Integrated the Google Maps API for address autocomplete, delivery-zone validation, distance-based fee calculation and live rider location on a customer-facing map.
- Handled payments with Stripe and cash on delivery, plus automated invoice generation and a refund flow for cancelled orders.
- Built an analytics dashboard using MongoDB aggregations for revenue trends, peak-hour analysis, best-selling items and average preparation time.

Skills

Frontend	React JS, Next JS, Redux Toolkit, RTK Query, React Router, HTML5, CSS3, Sass, Tailwind, Bootstrap
Backend	Node.js, Express.js, REST APIs, JWT Authentication, Socket.IO, Mongoose ODM
Databases	MongoDB, MySQL, Redis
Programming	JavaScript (ES6+), TypeScript, Python, C++, Java
Tools & DevOps	Git, GitHub, GitLab, Postman, Docker, NPM, Vite, Webpack, CI/CD
Services	Stripe, Cloudinary, Nodemailer, Google Maps API, OpenAI API
Concepts	MVC Architecture, Data Modeling, Caching, Testing, Agile
Languages	English, Arabic

Education

Sinai University	Ismailia, Egypt
B.S. IN COMPUTER SCIENCE AND ENGINEERING	2022 - 2026
• Algorithms, Data Structures, Complexity Analysis, System Design, Databases, Agile Project Management, Compilers, OOP, Networking, Operating Systems, C/Java, Computer Architecture, UML, ... etc.	